# A Firm-Level Database of Crypto-Exposed Firms[☆]

Ambrogio Cesa-Bianchi[†]
Bank of England
CEPR and CfM

Marco Pinchetti[‡]
Banque de France

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**Abstract**

This document describes the construction of a firm-level narrative dataset used to study the transmission of crypto-currency shocks. We start from a panel of publicly listed firms with earnings call transcripts and construct a measure of crypto exposure based on the frequency of crypto-related discussions. We then focus on the 200 firms with the highest average exposure over the sample period. We complement this measure with a LLM-based narrative classification of firms based on their business model and their economic linkages to crypto markets. Firms are assigned weights for each category based on a systematic review of earnings calls and firm descriptions. The resulting dataset contains firm-level exposure measures, business model classifications, and summary characteristics by group.

[☆]The full firm-by-firm documentation, including the LLM classification output, is available at `https://ambropo.github.io/CryptoShocks/index.html`. The views expressed in this paper are those of the authors, and not necessarily those of the Bank of England, the Banque de France, the Eurosystem, or their committees.

[†]Email: `ambrogio.cesa-bianchi@bankofengland.co.uk` Web: `sites.google.com/site/ambropo`

[‡]Email: `marco.pinchetti@banque-france.fr` Web: `sites.google.com/view/marcopinchetti`

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# 1 Overview

This document describes the construction of the firm-level narrative dataset used in the main paper, Cesa-Bianchi and Pinchetti (2026). The objective is to characterize how cryptocurrency shocks propagate across firms, and to map observed heterogeneity in equity price responses into economically meaningful transmission mechanisms.

The central challenge is that firm-level exposure to crypto markets is not directly observable. While text-based measures derived from earnings calls provide a scalable way to quantify the intensity of crypto-related discussions, they do not by themselves reveal the underlying economic channel through which firms are affected. Similar levels of measured exposure may reflect fundamentally different linkages to crypto markets—for example, direct balance sheet holdings, operational dependence on crypto activity, or exposure to the behavior of crypto-related clients. Distinguishing between these mechanisms is essential for interpreting the cross-sectional effects documented in the main analysis.

We address this problem by combining a quantitative measure of exposure with a structured narrative classification of firms. In a first step, we construct a firm-level measure of crypto exposure using earnings call transcripts and a predefined set of crypto-related keywords. This measure captures the intensity with which firms discuss crypto-related topics and provides a consistent ranking across a broad cross-section of firms. In a second step, we focus on the subset of firms for which this exposure is economically meaningful. Specifically, we restrict attention to the 200 firms with the highest average exposure over the sample period, thereby concentrating on firms where crypto developments are most likely to affect firm outcomes.

In a third step, we assign each firm in this subset to a business model group based on a detailed narrative assessment of its activities and its economic linkage to crypto markets. This classification maps the reduced-form exposure measure into a set of interpretable transmission channels, such as balance sheet effects, operating leverage, and demand-driven effects. The role of the narrative layer is therefore not to replace the quantitative measure, but to discipline its interpretation by linking it to underlying economic mechanisms.

Our approach is related to the broader narrative tradition in empirical macroeconomics, which uses qualitative information to identify or interpret economic shocks (e.g., Friedman and Schwartz (1963), Romer and Romer (2004)). In our setting, narrative evidence is used to interpret cross-sectional heterogeneity rather than to identify time-series shocks. This distinction is important: while the identification of crypto surprises is addressed separately in the event-level appendix, the present dataset provides the cross-sectional structure needed

to study how those shocks propagate across firms.

This appendix focuses exclusively on the construction of the firm-level dataset and its narrative classification. The resulting dataset complements the event-level analysis by providing a structured framework to analyze heterogeneity in transmission across firms with different types of exposure.

The appendix should be read together with the other components of the submission package. The file `event_instructions.md` contains the detailed instructions used to classify events, and the files `event_report.csv` and `event_dataset.html` provide, respectively, the structured output and a human-readable version of the classification results.

# 2 From Exposure Measure to Firm Selection

The starting point for the firm-level narrative dataset is the broad panel of publicly listed firms for which earnings call transcripts are available and for which the crypto exposure measure can be constructed. As described in the main text, this measure is based on the share of sentences in earnings calls that contain at least one crypto-related n-gram. Aggregating this information across calls and quarters yields a firm-level exposure index, $\text{CryptoExpo}_i$, which provides a consistent ranking of firms by the intensity of their crypto-related discussions.

We use this measure as a first-stage selection device. The objective is not to identify transmission mechanisms directly from the exposure measure itself, but rather to use it to isolate the subset of firms for which crypto-related developments are most likely to be economically relevant. We therefore rank all firms in the baseline sample by their average exposure, $\text{CryptoExpo}_i$, over the sample period and retain the 200 firms with the highest values.

This restriction serves a signal-extraction purpose. In the full cross-section, many firms exhibit only limited or sporadic references to crypto-related topics, making it difficult to distinguish economically meaningful exposure from background noise. Focusing on the upper tail of the exposure distribution allows us to concentrate on firms for which crypto-related discussions are persistent and quantitatively important, and therefore more likely to reflect genuine underlying linkages to crypto markets.

Table 1 reports summary statistics comparing the full baseline sample and the top-200 subsample. The differences in exposure levels are economically large. The mean exposure increases from 0.41 in the baseline sample to 2.74 in the top-200 sample, while the median rises from 0.03 to 0.73. This implies that the selection does not simply pick up a few extreme

outliers, but shifts the entire distribution upward: even the median firm in the top-200 sample exhibits substantially higher exposure than the average firm in the baseline sample. At the same time, the large gap between the mean and the median in both samples—particularly in the top-200—confirms the strong right-skewness already visible in Figure 1.

Importantly, this increase in exposure is not achieved by collapsing the cross-sectional diversity of the sample. The number of countries declines from 59 to 26 and the number of sectors from 11 to 10, indicating some concentration, but the top-200 firms remain broadly distributed across both geographic regions and industries. In other words, the restriction sharpens the signal along the exposure dimension without reducing the sample to a narrow set of firms. This balance is crucial for the subsequent narrative classification: it ensures that the analysis focuses on firms with meaningful crypto exposure while preserving sufficient heterogeneity to study different transmission mechanisms across sectors and countries.

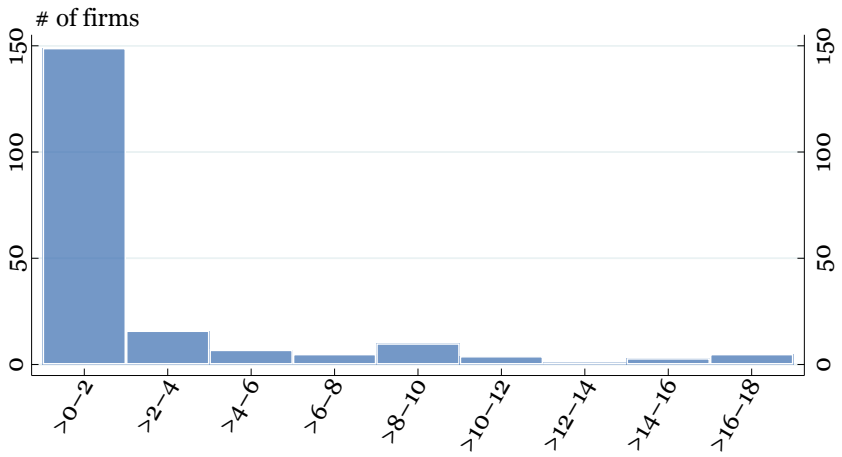
**Table 1** Baseline Sample and Top-200 Sample

| **Statistic** | **Baseline Sample** | **Top-200 Sample** |
|---|---|---|
| Number of firms | 1486 | 200 |
| Mean $\text{CryptoExpo}_i$ | 0.41 | 2.74 |
| Median $\text{CryptoExpo}_i$ | 0.03 | 0.73 |
| Number of countries | 59 | 26 |
| Number of sectors | 11 | 10 |

Note. The table compares the full baseline sample of firms with non-zero crypto exposure and the top-200 firms selected on the basis of the firm-level average exposure measure, $\text{CryptoExpo}_i$. The exposure measure is computed from earnings call transcripts as described in the main text. Sample period: 2017Q1–2024Q4.

Figure 1 shows the distribution of the firm-level exposure measure for the selected subsample. Two features stand out. First, the distribution remains highly right-skewed even within the top-200 firms, indicating that exposure is not only concentrated relative to the full sample, but also unevenly distributed within the most exposed group itself. A small number of firms exhibit extremely high values of $\text{CryptoExpo}_i$, while the majority of the top-200 still lie closer to the lower end of this restricted distribution. Second, the figure highlights the sharp separation between the bulk of firms in the baseline sample—clustered near zero—and the upper tail captured by the selection rule. This confirms that the top-200 cutoff effectively isolates a distinct subset of firms for which crypto-related discussion is systematically salient rather than incidental.

**Figure 1** Distribution of Crypto Exposure Across Top-200 Firms




Note. The figure shows the distribution of the firm-level crypto exposure measure, $\text{CryptoExpo}_i$, for the 200 firms with the highest average exposure in the baseline sample. Sample period: 2017Q1–2024Q4.

# 3 Narrative Classification of Firms

This section describes the narrative classification of the top-200 firms into crypto-related business models and transmission mechanisms. The classification is implemented using a large language model (LLM) that processes firm-level textual evidence under a fully specified and reproducible set of instructions. The purpose is to identify how cryptocurrency markets are economically relevant for each firm, based on the interpretation of earnings call transcripts, regulatory disclosures, and financial news.

In this sense, the exercise is the firm-level analogue of the event classification developed in the event-level appendix: rather than classifying news events by type, we classify firms by the economic mechanism through which crypto shocks are likely to affect their valuations. The use of an LLM allows us to scale this narrative classification in a consistent and transparent way, ensuring that the same decision rules are applied uniformly across firms while preserving a structured, evidence-based interpretation of each firm's business model.

## 3.1 Classification Methodology

The classification is implemented using a *large language model (LLM)* operating under a fully specified and reproducible set of instructions. Rather than relying on manual reading or keyword-based rules, the procedure delegates the interpretation of firm-level evidence to the LLM, which follows a structured decision protocol designed to recover the economic

mechanism through which crypto-related developments affect firm outcomes. The full set of instructions governing this process is fixed ex ante and applied uniformly across all firms, ensuring consistency and scalability of the classification. :contentReference[oaicite:0]index=0

For each firm, the LLM constructs a standardized evidence base through targeted retrieval over the period January 2017 to December 2024. The procedure combines three source types: earnings call transcripts, regulatory filings, and financial news coverage. Earnings call transcripts are the primary input, as they contain direct statements by management regarding strategy, revenues, risks, and business lines. The LLM retrieves multiple transcripts when available—typically spanning several years—to capture the persistence and evolution of crypto exposure. For firms with limited transcript coverage, the procedure relies more heavily on filings and press releases, while explicitly recording the resulting reduction in certainty. Under the protocol, at least one transcript quote should anchor every non-residual classification; if no such quote can be retrieved, the firm can still be classified, but the certainty score is capped.

In a second step, the LLM augments the transcript evidence with information from filings and financial news. Filings (e.g., 10-K reports) are used to quantify the materiality of crypto exposure, including revenue shares, custody balances, mining operations, or digital asset holdings. Financial news sources serve a complementary role by corroborating the classification and by identifying whether the firm is perceived as a crypto proxy in financial markets. This information is used to construct an orthogonal `invbase` indicator capturing exposure through the investor base rather than through operations.

The core of the methodology is a structured *decision tree* embedded in the LLM instructions. The model is required to map the collected evidence into one or more of six mutually exclusive mechanism categories, based on economically interpretable criteria. Crucially, assignments must be grounded in *substantive evidence*: the LLM is explicitly instructed to disregard incidental mentions of cryptocurrencies and to focus instead on documented linkages between crypto developments and firm revenues, costs, assets, or strategic positioning. This ensures that the classification reflects economic exposure rather than textual coincidence.

The LLM produces fractional assignments across categories, with weights expressed in increments of 0.1 and constrained to sum to one. These weights are determined by a combination of factors specified in the instructions: the frequency of references across sources, the economic materiality of each activity, and the emphasis placed on it by management. This structured weighting replaces discretionary judgment with a transparent and replicable rule. Firms with no meaningful or credible crypto exposure are assigned fully to the residual / unclassified category.

In addition to mechanism weights, the LLM records two ordinal measures of materiality. `RevShare` captures the approximate share of firm revenue attributable to crypto-related activities, while `AssetShare` captures the importance of crypto-related assets on the balance sheet. These measures are extracted from the same evidence base and provide a compact summary of the economic scale of exposure, without directly influencing the categorical assignment.

Finally, each firm is assigned a certainty score from 1 to 4. This score is generated by the LLM based on the strength, consistency, and coverage of the underlying evidence. A score of 4 corresponds to multiple strong sources and a clear dominant mechanism, while lower scores reflect weaker or more ambiguous evidence, limited source availability, or reliance on indirect inference. Importantly, the scoring criteria are themselves embedded in the instructions, ensuring that uncertainty is measured consistently across firms.

Taken together, this LLM-based framework transforms qualitative firm-level information into a structured and reproducible dataset. By combining standardized inputs, explicit decision rules, and constrained outputs, the procedure delivers a classification that is both economically interpretable and scalable across a large cross-section of firms.

## 3.2 Structure of the Firm-Level Outputs

The classification procedure produces two distinct but complementary outputs: (i) a collection of firm-by-firm narrative reports, and (ii) a structured dataset in CSV format. Both outputs are generated from the same underlying classification protocol and contain identical core information, but they are designed for different purposes.

The first output is a *firm-level narrative report*. Each firm in the top-200 sample is associated with a standalone, human-readable document that records the full reasoning underlying the classification. These reports are designed to ensure transparency and traceability. Each report contains four main components. First, a summary block reports the firm identifier, rank in the top-200 distribution, exposure score, mechanism weights, certainty score, investor-base indicator, and materiality measures. Second, a concise narrative overview describes the firm's business model and explains how crypto-related activities fit within it. Third, a classification section provides a structured justification for the assignment of weights across mechanisms. Fourth, an evidence section lists the underlying sources, including earnings call excerpts, regulatory filings, and financial news references. Taken together, these reports document how qualitative information is translated into a quantitative representation for each firm.

The second output is a *structured dataset in CSV format*. This dataset provides a machine-readable representation of the same classification results and is used directly in the empirical analysis. Each row corresponds to a firm, and each column corresponds to a variable derived from the narrative classification procedure, including mechanism weights, certainty scores, investor-base indicators, and measures of economic materiality. Relative to the narrative reports, the CSV dataset abstracts from the underlying textual evidence and retains only the standardized quantitative outputs required for econometric analysis.

Table 2 summarizes the structure of the firm-level outputs and the variables included in the dataset. The table should be read as a bridge between the narrative reports and the CSV dataset: it documents how the qualitative information contained in the reports is mapped into the structured variables used in the empirical analysis.

**Table 2** Structure of Firm-Level Reports

| Component | Description |
|---|---|
| Evidence | Compilation of verbatim excerpts from earnings call transcripts, SEC filings, and financial news. Documents the factual basis for identifying crypto-related activities. |
| Firm Overview | Summary of the firm's core business model and the channels through which crypto-related activity may affect operations or revenues. |
| Exposure Classification | Assignment of the firm to a primary transmission mechanism (e.g., infrastructure, financial services, asset exposure), based on a structured interpretation of the evidence. |
| Weight Assignment | Translation of qualitative classification into quantitative exposure measures, including indicators for revenue-based exposure and balance sheet exposure. |
| Ambiguities / Limitations | Discussion of potential measurement issues, including overlapping channels, indirect exposure, or limitations in disclosure. |

Together, the two outputs form a complete and internally consistent classification pipeline. The narrative reports provide detailed documentation and allow for independent verification of each classification decision, while the CSV dataset ensures that the same information can be used systematically in the quantitative analysis.

## 3.3 Group Definition and Statistics

We classify each firm in the top-200 sample into one of six broad groups reflecting the dominant transmission mechanism through which cryptocurrency developments may affect

firm value. The relevant groups are: *crypto holders*, *crypto miners*, *crypto financial services*, *crypto non-financial services*, *crypto infrastructure providers*, and a *residual / unclassified* category. The first five groups correspond to economically interpretable channels, while the last category captures firms for which the available evidence does not support a sufficiently sharp assignment.

The classification is based on the firm's *dominant business model*. Crypto holders are firms whose exposure arises primarily through balance-sheet holdings of digital assets. Crypto miners are firms whose revenues depend directly on mining activity. Crypto financial services firms provide exchange, brokerage, custody, payments, or crypto investment products. Crypto non-financial services firms provide services to the crypto ecosystem without directly intermediating or holding digital assets, such as consulting, software integration, or platform services. Crypto infrastructure providers supply the hardware, compute, hosting, or energy inputs used by the crypto sector. Finally, firms are assigned to the residual / unclassified group when the evidence points only to incidental crypto references, or when the available information is too weak or too mixed to support a meaningful classification.

Although the objective is to identify a dominant mechanism, some firms operate across multiple crypto-related activities. The classification therefore allows for *multi-activity firms*. In these cases, weights can be distributed across categories in increments of 0.1, with the full set of mechanism weights summing to one. This weighting scheme is used to reflect differences in economic importance across activities while preserving comparability across firms. In practice, however, most firms load overwhelmingly on a single category, so the resulting classification is close to binary for the large majority of observations.

Two decision rules are important. First, the assignment is based on *substantive* evidence rather than passing references to cryptocurrencies. A firm is not classified as crypto-related merely because management mentions Bitcoin or blockchain in an earnings call. The relevant criterion is whether the source material documents a genuine economic linkage between firm outcomes and the crypto sector. Second, when multiple categories are potentially relevant, ties are broken by economic materiality: the dominant category is the one most closely tied to revenues, assets, or the firm's strategic positioning over the sample period. When no category clearly dominates, the residual / unclassified group is used.

Table 3 summarizes the classification taxonomy.

Figure 2 reports the distribution of firms across transmission mechanisms within the top-200 sample. Two features stand out. First, the cross-sectional allocation is economically meaningful and far from uniform: a substantial share of firms is concentrated in infrastructure, financial services, and non-financial services, while relatively fewer firms are classified

**Table 3** Classification of Firms by Dominant Transmission Mechanism

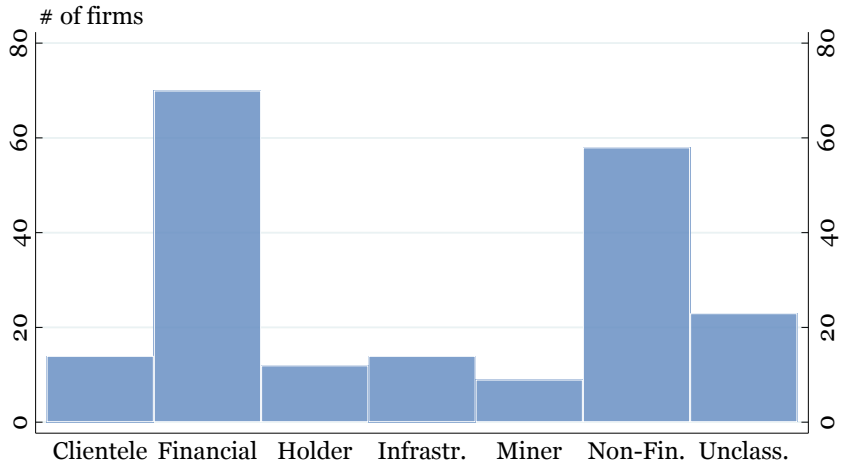
| Category | Description |
|---|---|
| Crypto Holders | Firms whose exposure arises primarily from balance-sheet holdings of cryptocurrencies. Their valuation is directly affected by price fluctuations through marked-to-market gains and losses. Representative firms include MicroStrategy and Tesla. |
| Crypto Miners | Firms engaged in the validation of blockchain transactions, whose revenues depend on block rewards and transaction fees. Their profitability is tightly linked to crypto prices and energy costs. Representative firms include Marathon Digital and Hut 8. |
| Crypto Financial Services | Firms providing trading, brokerage, custody, or other intermediation services related to cryptocurrencies. Their exposure is driven by trading volumes, user activity, and market participation. Representative firms include Coinbase and Robinhood. |
| Crypto Non-Financial Services | Firms integrating crypto technologies into their products or platforms, such as payments, wallets, or blockchain-based applications. Exposure reflects adoption dynamics and demand for crypto-related services. Representative firms include Block and PayPal. |
| Crypto Infrastructure | Firms supplying hardware, software, or cloud infrastructure that supports the crypto ecosystem. Their exposure operates indirectly through demand for inputs used in mining or blockchain operations. Representative firms include NVIDIA and AMD. |
| Unclassified | Firms for which available information does not support a clear identification of a dominant transmission mechanism, either due to limited disclosure or overlapping channels of exposure. |

Note. Each firm is assigned to the category corresponding to its dominant economic linkage to cryptocurrency markets, based on the narrative classification procedure described in Section **??**. While the underlying framework allows firms to load on multiple categories, the dominant-category assignment is used in the main analysis. The *Unclassified* category includes firms for which the available evidence does not support a sufficiently precise attribution to a dominant mechanism.

as pure crypto holders or miners. This pattern reflects the broader industrial organization of the crypto ecosystem, in which indirect exposure through services and inputs is more prevalent than direct balance-sheet or mining exposure. Second, the primary and weighted counts are nearly identical across all groups. This indicates that, despite allowing for multi-activity firms, the dominant-mechanism classification is effectively discrete in practice: most

firms are assigned a weight close to one in a single category, with only limited overlap across mechanisms.

**Figure 2** Number of Firms by Transmission Mechanism (Top-200 Sample)

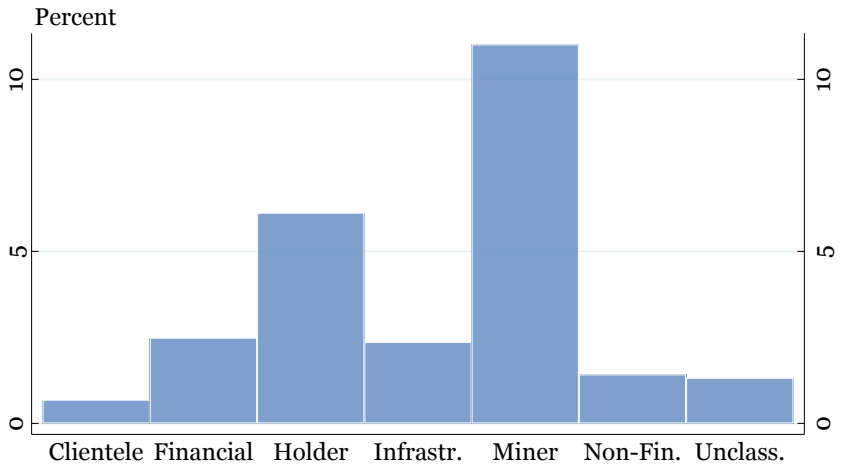



Note. Number of firms by business model group among the 200 most crypto-exposed firms. Each group shows two bars: primary count (firms assigned to the group by dominant weight) and weighted count (sum of business model weights across all firms). Near-identical bars indicate that business model classification is de facto binary. Sample period: 2017Q1–2022Q4.

Figure 3 complements this evidence by reporting the average level of crypto exposure within each transmission mechanism. The results reveal substantial heterogeneity across groups. Firms classified as crypto financial services and crypto miners exhibit the highest average exposure, consistent with business models that are directly tied to trading activity, user participation, or mining revenues. Crypto holders also display elevated exposure, reflecting the prominence of balance-sheet positions in shaping firm-level outcomes. By contrast, crypto infrastructure providers and non-financial service firms tend to exhibit lower average exposure, consistent with a more indirect linkage operating through input demand or complementary services.

Comparing primary and weighted averages further reinforces the interpretation that most firms are cleanly assigned to a single mechanism. The close alignment of the two measures indicates that allowing for fractional weights does not materially affect group-level averages, again suggesting that multi-activity firms are relatively rare or that secondary activities are economically minor. Taken together, the two figures show that the classification captures both meaningful cross-sectional variation in business models and systematic differences in the intensity of crypto exposure across those models.

**Figure 3** Average Crypto Exposure by Transmission Mechanism (Top-200 Sample)




Note. Average crypto exposure ($\overline{\text{CryptoExpo}}_i$) by business model group among the 200 most crypto-exposed firms. Each group shows two bars: primary average (mean exposure of firms assigned to the group by dominant weight) and weighted average (exposure weighted by business model weights). Sample period: 2017Q1–2022Q4.

## 3.4 Illustrative Example: NVIDIA

To illustrate the structure and content of the firm-level outputs, this subsection presents the classification report for NVIDIA. The example shows how the qualitative evidence is organized in the firm-by-firm report and how the same information maps into the structured dataset summarized in Table 2. Figure 4 reproduces the NVIDIA report in full. The first three panels contain the summary block, the firm overview, and the discussion of exposure channels. The final two panels contain the weight assignment and the detailed evidence section. Together, these panels illustrate how the classification procedure moves from source material to mechanism weights, certainty measures, and materiality indicators in a transparent and reproducible way.

**Weight Assignment.** The report assigns NVIDIA fully to the *infrastructure* category, reflecting a clear and internally consistent narrative across all sources. The core evidence indicates that the firm's crypto exposure operates through the supply of mining-related hardware—specifically GPUs and dedicated crypto-mining processors (CMPs)—rather than through any direct participation in crypto markets. Earnings call transcripts repeatedly document that management explicitly linked fluctuations in GPU demand to cryptomining activity and disclosed revenues from crypto-specific products. Regulatory filings further confirm that cryptocurrency demand materially affected certain business segments, while financial news coverage consistently interprets NVIDIA as a key upstream supplier to the

mining ecosystem.

The classification rules out alternative channels on the basis of explicit negative evidence. There is no indication that NVIDIA held crypto assets on its balance sheet (*holders*), operated mining facilities (*miners*), or provided exchange, custody, or brokerage services (*financial services*). The *business services* category is also not supported, as the relevant exposure does not arise from advisory or software services but from the sale of physical inputs into the crypto production process. The dominance of the infrastructure channel is therefore not the result of relative weighting across competing mechanisms, but of a qualitatively unambiguous mapping from the evidence to a single economic linkage.

The report also records a high certainty score (4), reflecting the consistency of the narrative across independent sources and over time. The materiality measures further refine the interpretation of exposure. The revenue-based indicator (`RevShare`) is set to one, as the firm reported identifiable crypto-related revenues—such as CMP sales—yet these remain small relative to total revenues, implying that crypto is not a primary driver of firm-wide performance. The asset-based indicator (`AssetShare`) is set to zero, consistent with the absence of crypto assets on the balance sheet. Finally, the positive investor-base indicator (`invbase`=1) reflects the fact that NVIDIA is perceived by market participants as a crypto-sensitive stock, as evidenced by financial media coverage linking its equity performance to developments in the crypto cycle.

**Ambiguities and limitations.** While the overall classification is unambiguous, the report explicitly considers potential boundary cases. The main ambiguity concerns whether NVIDIA's activities could partially fall under *business services*, given its role in supplying general-purpose computing hardware that is also used in non-crypto applications. This interpretation is rejected because the relevant exposure operates through the provision of standardized hardware inputs rather than customized services or advisory functions. A second potential ambiguity relates to the distinction between infrastructure provision and indirect participation in mining. The evidence consistently supports the former: NVIDIA does not operate mining facilities or directly validate transactions, but instead supplies inputs to third-party miners. More broadly, the classification is subject to the general limitations of text-based evidence. Earnings call disclosures may emphasize particular business segments at specific points in time, and financial news coverage may reflect market narratives rather than purely fundamental linkages. These limitations are mitigated by triangulating across multiple independent sources and by requiring consistency of evidence over time.

**Evidence base and sources.** The report draws on three primary categories of source material, which are systematically documented in the evidence section of the firm-level out-

put. First, earnings call transcripts provide direct statements by management regarding the role of crypto-related demand in shaping product sales, pricing dynamics, and revenue composition. Second, regulatory filings (including annual and quarterly reports) provide audited and standardized disclosures on segment revenues, risk factors, and the economic relevance of cryptocurrency-related demand. Third, financial news coverage offers an external perspective on how the firm is interpreted by market participants, including the extent to which its equity is viewed as a proxy for crypto market developments. The evidence section reports specific excerpts from these sources, together with precise references, allowing the reader to trace each element of the classification—mechanism weights, certainty scores, and materiality indicators—back to the underlying documentation. This structured aggregation of heterogeneous sources is a key feature of the methodology, ensuring both transparency and reproducibility of the firm-level classifications.

## NVIDIA

- Rank: 87 | Expo: 0.89 | id_str: US67066G1040
- Weight summary: holders 0.0, miners 0.0, finserv 0.0, bizserv 0.0, infrastr 1.0, unclass 0.0
- Investor base amplifier (invbase): 1
- Certainty: 4
- RevShare: 1 | AssetShare: 0
- Primary sources: earnings call transcripts (2018, 2021, 2022), SEC filings (2019 Form 10-K, 2018 Form 10-Qs), SEC enforcement release, financial news

### Firm Overview

NVIDIA is a U.S.-listed semiconductor and computing platform company whose core businesses span gaming GPUs, data-center accelerators, professional visualization, and automotive computing. Within the January 2017 to December 2024 sample period, its crypto exposure came through supplying GPU hardware specifically used for cryptocurrency mining and, at times, dedicated crypto-mining processors rather than through directly holding, mining, or transacting in digital assets. (sec.gov)

### Exposure Channels Identified

The relevant mechanism is `infrastr`. Management explicitly discussed crypto-specific GPUs/CMP products, stated that mining demand affected GPU demand, and reported dedicated CMP revenue in earnings calls. (fool.com)

`holders`, `miners`, and `finserv` do not fit. The evidence shows NVIDIA sold chips and related hardware into the mining ecosystem, but there is no evidence in the sample period that it held crypto as treasury, operated mining rigs itself, or directly provided exchange, custody, brokerage, or other crypto financial services. (sec.gov)

`bizserv` is also less appropriate than `infrastr` because the exposure was not consulting or software support around crypto; it was hardware supply for mining workloads. The SEC's 6 May 2022 release further confirms cryptomining was a significant driver of gaming-related demand in parts of fiscal 2018, reinforcing that the mechanism was infrastructure provision to the crypto ecosystem. (sec.gov)

### Weight Assignment

A full `infrastr = 1.0` assignment is appropriate because all substantive crypto references point to NVIDIA's role as a supplier of mining-capable GPUs and dedicated CMP products. There is no support for splitting weight into other categories: crypto exposure affected revenue through hardware demand from miners, not through owned crypto assets or direct crypto intermediation. `RevShare` is set to 1 because disclosed crypto-specific revenue was real but peripheral relative to total company revenue—for example, fiscal 2019 cryptocurrency-specific product revenue was $306 million against total revenue of $11.716 billion, and Q1 FY2022 CMP revenue was $155 million. `AssetShare` is set to 0 because no evidence indicates material crypto assets on NVIDIA's balance sheet. (sec.gov)

### Ambiguities / Limitations

The main ambiguity is that some miner demand flowed through general-purpose GeForce gaming GPUs, which NVIDIA said was hard to quantify precisely, so the exact crypto share of gaming revenue was not always cleanly segmented. However, classification is still unambiguous because management repeatedly referenced mining demand and dedicated CMP products, the 2019 Form 10-K disclosed cryptocurrency-specific product revenue, and the SEC formally stated that cryptomining materially affected the gaming business in fiscal 2018. (fool.com)

**Figure 4** Firm Report Example: NVIDIA (Overview)

Note. Panels reproduced from the standardized firm-level report for NVIDIA. The first three panels report the summary block, the firm overview, and the exposure-channel discussion.

## Evidence

**Earnings calls:**

- 16 May 2018, Q1 FY2019 earnings call transcript: *"Cryptocurrency demand was, again, stronger than expected"* and crypto-specific GPUs in OEM were $289 million. — The Motley Fool transcript
- 16 May 2018, Q1 FY2019 earnings call transcript: Jensen Huang said NVIDIA's strategy was to create a SKU so "crypto miners" could fulfill demand that way. — The Motley Fool transcript
- 26 May 2021, Q1 FY2022 earnings call transcript: management said gaming also benefited from crypto-mining demand and that CMP revenue was $155 million in Q1, with Q2 CMP sales expected at $400 million. — The Motley Fool transcript
- 16 February 2022, Q4 FY2022 earnings call transcript: management said GPUs were capable of cryptocurrency mining, noted limited visibility on mining-driven GPU demand, and disclosed CMP revenue of $24 million. — The Motley Fool transcript

**Filings (10-K / press releases):**

- 21 February 2019, 2019 Form 10-K: NVIDIA disclosed revenue from cryptocurrency-specific products of $306 million in fiscal 2019 and $273 million in fiscal 2018. — SEC 2019 Form 10-K
- 21 February 2019, 2019 Form 10-K: the company said demand for desktop gaming GPUs used for cryptocurrency mining and its after-effects distorted gaming revenue trends. — SEC 2019 Form 10-K
- 16 August 2018, Q2 FY2019 Form 10-Q: GPU business results were led by gaming, professional visualization, and datacenter, offset by a substantial decline in cryptocurrency GPUs. — SEC Q2 FY2019 Form 10-Q

**Financial news / other:**

- 6 May 2022, SEC enforcement release: the SEC said NVIDIA failed to disclose that material growth in gaming revenue in parts of fiscal 2018 was driven in significant part by cryptomining. — SEC press release
- 9 February 2018, CNBC: NVIDIA shares surged after earnings as analysts highlighted strong GPU demand from the cryptocurrency mining market. — CNBC
- 16 November 2018, CNBC: financial media described NVIDIA as suffering a "crypto hangover" as mining demand evaporated, indicating direct crypto-sentiment spillover into the stock. — CNBC
- 17 November 2021, CNBC: the outlet reported NVIDIA's crypto-mining chip sales dropped sharply and quoted management on limited visibility into mining-driven GPU demand. — CNBC

**Figure 4** Firm Report Example: NVIDIA (Evidence)

Note. Continued. The final two panels report the weight assignment and the detailed evidence section, including excerpts from earnings calls, regulatory filings, and financial news sources.

# References

Cesa-Bianchi, A. and M. Pinchetti (2026): "Crypto Shocks," Working paper.

Friedman, M. and A. J. Schwartz (1963): *A Monetary History of the United States, 1867–1960*, Princeton, NJ: Princeton University Press.

Romer, C. D. and D. H. Romer (2004): "A New Measure of Monetary Shocks: Derivation and Implications," *American Economic Review*, 94, 1055–1084.

# LLM Instructions

This section reproduces, verbatim, the complete set of instructions supplied to the language model for the narrative classification of firms described in Section 3. The instructions were provided to the model at the start of each session and remained fixed throughout the project. Their purpose is to ensure that the classification of firms into business models is conducted in a systematic, reproducible, and economically disciplined manner.

**Crypto Firms Classification Instructions**

I have some instructions for a project on ‘‘crypto exposure’’. The objective is to classify listed firms based on the economic channels through which developments in cryptocurrency markets may affect their equity prices. The classification focuses on mechanisms rather than raw exposure.

**Input**

For each firm, I will provide a row from a CSV file containing firm identifiers and metadata (including name, sector, and rank). No documents are uploaded. All relevant information must be retrieved through targeted web searches and direct scraping of source pages.

The variable expo_firm is provided for bookkeeping but should not be used to identify or disambiguate firms.

**Goal**

For each firm, assign fractional weights across six categories that capture distinct transmission mechanisms:

- holders
- miners
- finserv
- bizserv
- infrastr
- unclass

The weights must sum to 1.0 and be expressed in increments of 0.1. In addition, assign a binary indicator capturing whether the firm is subject to a crypto-related investor base (invbase).

**Step-by-Step Analysis Process**

**1. Retrieve Earnings Call Transcripts**

Search for and extract earnings call transcripts over the period 2017--2024. Prioritize sources such as Seeking Alpha, Motley Fool, and investor relations

pages. Extract all crypto-related passages and record short quotes as evidence. At least one transcript-based quote must be identified to anchor the classification.

**2. Retrieve Supporting Evidence**

Complement transcript evidence with firm filings (e.g. 10-K reports) and financial news sources. These sources are used to corroborate the presence, nature, and materiality of crypto-related activities. All searches are firm-specific and cover the full sample period.

**3. Apply Decision Tree Classification**

Assign mechanism weights based only on explicitly documented activities:

- **holders**: firms holding crypto assets on balance sheet for investment purposes
- **miners**: firms engaged in crypto mining activities
- **finserv**: firms providing crypto-related financial services (trading, custody, issuance)
- **bizserv**: firms providing non-financial services to the crypto ecosystem
- **infrastr**: firms supplying hardware, compute, or energy infrastructure to crypto activities
- **unclass**: no meaningful crypto exposure identified

Classification must reflect the dominant business model. If multiple activities are present, weights should be distributed according to their relative economic importance, based on frequency of mentions, management emphasis, and materiality in filings.

**Weighting Rules**

- Weights must sum to 1.0
- Use increments of 0.1 only
- Assign a single category weight of 1.0 if evidence clearly supports one dominant mechanism
- Use multiple categories only when supported by substantive and consistent evidence
- Assign unclass = 1.0 if no credible crypto exposure is identified

**Investor Base Amplifier**

Assign invbase = 1 if there is explicit evidence that the firm is treated by investors as a crypto proxy or is subject to crypto-driven retail flows. Otherwise assign 0. This variable is orthogonal to mechanism weights.

**Certainty Score**

Assign a certainty score from 1 to 4:

- 4: Multiple strong sources, explicit and material crypto activity
- 3: Clear evidence with supporting corroboration
- 2: Limited or indirect evidence
- 1: Weak or inferred classification

Assign NA if the firm is unclassified.

**Materiality Measures**

Record two ordinal measures:

- Revenue share (crypto-related revenue as share of total)
- Asset share (crypto assets as share of total assets)

Each measure is coded on a scale from 0 (negligible) to 5 (dominant), or NA if not determinable.

**Reporting Format**

For each firm, produce:

- A structured markdown report summarizing classification, evidence, and ambiguities
- A CSV row containing mechanism weights, certainty score, investor base indicator, and materiality measures

**Practical Rules**

- Each firm is treated independently; no carryover information across firms
- All classifications must be based on retrieved evidence, not prior knowledge
- Stop searching once classification is stable
- Always write in the third person